

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083)

TIME ALLOWED : 1 Hr. 45 Min.

General Instructions:

1. The paper is divided into 5 Sections - A, B, C, D and E.
2. All questions are compulsory.
3. All programming questions are to be answered using Python Language only.
4. Write all the sub-parts of a question together sequentially.

SECTION - A

Q.1 Answer the following: (1 X 13 = 13)

- (a) Name the network device that connects two dissimilar networks working on different protocols.
- (b) Which protocol is used to transfer files from one computer to another over the Internet?
- (c) Name the network topology in which all the nodes are connected to a central hub or switch.
- (d) Which guided transmission medium carries data in the form of light signals and is immune to electromagnetic interference?
- (e) The bandwidth of a communication channel is measured in _____.
- (f) Name the Python module that must be imported to use the `dump()` and `load()` functions on a binary file.
- (g) Which method of a csv writer object is used to write several rows into a CSV file at once?
- (h) What is the default mode in which a file is opened by `open()` when no mode is specified?
- (i) Write a statement to place the file pointer of the file object `F` at the 20th byte from the beginning of the file.
- (j) What will be the output of the following code?

```
def f(a, b=2):  
    return a ** b  
print(f(3), f(2, 3))
```
- (k) A user-defined function that does not have a `return` statement returns _____ by default.
- (l) **Assertion (A):** A variable created inside a function cannot be accessed outside that function.
Reason (R): The scope of a local variable is limited to the function in which it is defined.
 - i. Both A and R are True and R is the correct explanation for A.
 - ii. Both A and R are True and R is not the correct explanation for A.
 - iii. A is True but R is False.
 - iv. A is False but R is True.

(m) **Assertion (A):** Opening an existing text file in 'a' mode does not delete its previous contents.

Reason (R): In 'a' mode the file pointer is placed at the end of the file.

- i. Both A and R are True and R is the correct explanation for A.
- ii. Both A and R are True and R is not the correct explanation for A.
- iii. A is True but R is False.
- iv. A is False but R is True.

SECTION - B

Q.2 Write the full form of the following: (2)

(a) SMTP (b) URL (c) XML (d) TCP/IP

Q.3 What will be the output of the following Python code? (2)

```
def modify(L, x):  
    L.append(x)  
    x = x * 2  
    L[0] = L[0] + x  
    return x
```

```
nums = [5, 10]  
val = 3  
r = modify(nums, val)  
print(nums, val, r)
```

Q.4 Differentiate between the `write()` and `writelines()` methods of a file object. Give one example of each. (2)

SECTION - C

Q.5 Differentiate between the following (one point each): (3)

- (a) Hub and Switch
- (b) Web Browser and Web Server
- (c) HTML and XML

Q.6 A text file "**POEM.TXT**" contains the following lines: (3)

```
Twinkle twinkle little star
How I wonder what you are
Up above the world so high
Like a diamond in the sky
```

Write the output of each of the following code segments:

- (a)

```
f = open("POEM.TXT", "r")
print(f.read(7))
f.close()
```
- (b)

```
f = open("POEM.TXT", "r")
L = f.readlines()
print(len(L), L[2].split()[0])
f.close()
```
- (c)

```
f = open("POEM.TXT", "r")
f.seek(8)
print(f.read(7))
print(f.tell())
f.close()
```

Q.7 Write a Python function **FREQ(L)** that accepts a list **L** of integers as argument and returns (3)
a dictionary in which each distinct element of the list is a key and its number of occurrences is the value.

For example, if **L = [4, 2, 4, 7, 2, 4]**, the function should return **{4: 3, 2: 2, 7: 1}**.

Also write a statement to call the function and display the returned dictionary.

SECTION - D

Q.8 A binary file "**STUDENT.DAT**" stores records of students, where each record is a list in (4)
the form **[AdmNo, Name, Percent]**. Write the following user-defined functions:

- (a) **addrec()** – to accept the details of a student from the user and append the record to the file "**STUDENT.DAT**".
- (b) **countrec()** – to read the file "**STUDENT.DAT**", display the details of those students whose Percent is above 75, and return the number of such students.

Q.9 Answer the following questions: (4)

- (a) What was ARPANET? Name the network that evolved from it in the mid-1980s for academic research.
- (b) Write one difference between Circuit Switching and Packet Switching.
- (c) Why is microwave transmission called line-of-sight transmission?
- (d) What is an IP address? Give one example.

Q.10 Write the output of the following code segments:

(2+2)

```
(a) p = 5
    def calc(q):
        global p
        p = p + q
        q = q + p
        return q
    r = calc(p)
    print(p, r)
    r = calc(r)
    print(p, r)

(b) def show(name, marks=40, grade='C'):
    if marks >= 80:
        grade = 'A'
    print(name, marks, grade, sep='-')
show('Riya', 85)
show('Amit', grade='B')
show(marks=90, name='Joy', grade='B')
show('Sam', 60, 'B')
```

SECTION - E

Q.11 Ananya is maintaining the records of a school library in a CSV file named **"LIBRARY.CSV"**. Each record has the fields [BookID, Title, Author, Qty]. **(5)**

Help her by answering the following:

(a) Name the module and the function used to create a writer object for a CSV file. **(1)**

(b) Write a function **AddBook()** that accepts the details of a book from the user and appends it as a record to "LIBRARY.CSV". **(2)**

(c) Write a function **ShowLowStock()** that reads "LIBRARY.CSV" and displays the details of all books whose Qty is less than 5, along with the total number of such books. **(2)**

- Q.12 Global Tech Solutions** is setting up a new campus in Durgapur with four blocks – **(5)**
ADMIN, TECH, HR and ACCOUNTS. Its head office is in Delhi. The distances between the blocks and the number of computers in each block are given below:

Blocks	Distance
ADMIN to TECH	60 m
ADMIN to HR	120 m
ADMIN to ACCOUNTS	90 m
TECH to HR	50 m
TECH to ACCOUNTS	110 m
HR to ACCOUNTS	70 m

Block	No. of Computers
ADMIN	25
TECH	120
HR	30
ACCOUNTS	20

- (a) Suggest the most suitable block to house the server of this organisation. Justify your answer.
- (b) Draw a cable layout to efficiently connect the various blocks within the Durgapur campus.
- (c) Suggest the placement of a Repeater in the layout drawn by you, with justification.
- (d) Which type of network (PAN / LAN / MAN / WAN) will be formed when the Durgapur campus is connected to the Delhi head office?
- (e) The company wants to make voice calls over the Internet between Durgapur and Delhi. Name the protocol/technology that can be used for this purpose.

***** End of Question Paper *****